

# Prediction of Battery Failure in EVs Using Machine Learning : A Case Study

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**Abstract** – Electric Vehicles have been revolutionary in the automobile industry as they provide a more sustainable method of transportation. The battery is the heart of an EV as it stores electricity, making it environment friendly. They play a very important role in the reliability and overall functionality of an EV. Most EVs use Lithium-Ion batteries nowadays. These batteries are preferred for their long cycle lives and high energy density. Considering its importance, it is crucial to understand the reason for and predict battery failure to ensure the user's safety and the system's cost-effectiveness. Battery failure can be detected by monitoring the voltage, current, State of Charge and Temperature. Using a dataset containing the required parameters, a Machine Learning algorithm can be deployed to detect any failure by identifying any anomaly in the battery's working. Various algorithms, such as K Nearest Neighbour, Support Vector Machine, and Random Forest algorithm, can be used to predict battery failure. This study aims to determine which algorithm is best suited for this model.

**Keywords** – SoC, Temperature, Support Vector Machine.

## I. INTRODUCTION

Electric vehicles have been the biggest advancement in the automobile industry in the past few years. They are vehicles based on electric batteries which run on the electricity stored in these batteries. Compared to internal combustion engines, EVs are more energy efficient and environmentally friendly. Though the initial investment is high, they have running costs much lower than IC engines.

One of the main components of an EV is its battery pack. These battery packs store electricity in chemical form and release it to power the motors when the vehicle runs. They are typically Lithium-Ion and can be recharged. These batteries are the heart of EVs, and the failure of these batteries can cause serious issues. Prediction of battery failure is crucial for the longevity and cost-effectiveness of an EV. Previous studies have shown a direct relationship between battery failure rates and the graph plotted between the State of Charge (SoC) and its open circuit voltage [1]. A method to analyse different types of failure was proposed to prevent such occurrences in the future [2].

Unscented Kalman Filter-based techniques using State of Health as their primary prediction parameter can also be used. SoC and internal resistance of the battery also play an important part in the analysis [3]. The Battery Management systems (BMS) of EVs can be equipped with ML for better accuracy, and IoT can be used to establish a network for remote monitoring and scheduling maintenance [4]. Changes in thermal energy during the discharge of the battery can also be used to predict failure in advance. Such techniques also use the hysteresis characteristics to detect degradation [5].

Inconsistencies in battery current, voltage and temperature are good indicators of battery failure. The AOM-ARIMA-LSTM method used those parameters to predict any anomaly in the battery's working [6]. The temperature of a battery is one of the most reliable parameters for predicting failure. Temperature variation has negative impacts on the parameters and overall life of the battery. Run-to-failure and preventive and predictive maintenance are some of the features AIOT-based systems provide. It also assists in fault detection and notification [7]. SoC is also one of the most reliable parameters for battery failure prediction. Though SoC cannot be measured directly, it can be calculated using different methods, such as Kalman Filter and Fuzzy Logic. Other data-driven approaches include deploying machine learning algorithms such as linear regression, random forest, XGBoost, etc., for estimating a battery's SoC. Previous studies have also made use of current through and voltage across a battery and temperature for SoC estimation. These data-driven methods offer better accuracy in determining the State of Charge and are more reliable in the long run [8-9, 12].

Electric Vehicles (EVs) mostly use Lithium-Ion (Li-On) batteries over lead acid batteries due to their reliability and enhanced safety. But Li-On batteries are highly sensitive to high temperatures and can lead to explosions. Therefore, proper BMS has to be designed to reduce the safety hazards while increasing the life of the battery [10]. The dynamic and non-linear behaviour of the battery can make the prediction of battery ageing a difficult task. Different regression models can be used for this prediction. Parameters such as current and

voltage profiles during charging and discharging are used in such analysis [11].

Previously, capacitors, resistors and battery cycling have been used to project a battery's degradation profile. There have also been studies where mathematical and physics-based models were used. However, these studies require more data to produce accurate results since batteries differ in chemical properties. Previous research also show the importance of various parameters and the need for a system for predicting battery failure. The model aims to use different regression algorithms to accurately and efficiently predict battery failure in advance, allowing the user to be more careful about the battery condition of the EV.

This paper consists of 6 sections, with introduction as section one. Section two presents "Related Work". Section three describes "Working of the proposed system". This section comprises block diagrams representing the proposed system and its working and brief explanations. Section four presents "Machine learning models", with information about every ML model used in this work. Section five provides "Results and Analysis". The outputs obtained for each ML model are mentioned and analysed in this section. Section sixth presents "Conclusion", where the inferences from this work are mentioned. The final section of this paper is "References", in which all the papers referred to are mentioned.

## II. RELATED WORK

The main challenges faced with respect to EVs are the thermal management and SoC prediction. The BMS of an EV is designed in such a way that all these challenges can be eliminated [14]. Different technologies have been developed in order to deal with thermal runaway and other factors which produce heat. Limiting the current to the battery and optimizing charge and discharge currents are some of the ways of preserving the health of the battery. This also helps in improving safety against fire-related accidents [15].

This work aims to analyse the performance of different Machine Learning models in predicting battery failure in Electric Vehicles. Existing technology have only made use of less data such as voltage, current and temperature to train the machine learning model. Various other parameters such as heating circuit data and environmental data should also be taken into consideration. Using the hysteresis characteristics, the rate of failure and Remaining Useful Health (RUH) can be predicted. Such methods are more viable for batteries having low voltage and power ratings and can predict failure around 6 months in advance. Different probabilistic method have also been proposed in the past. These can predict battery failure due to thermal runaway or over discharging.

Advanced Machine Learning models such as Optimizable Gaussian Process Regression (GPR) and Matern 5/2 GPR have been used in the past which provide a high accuracy when used on smaller datasets, in particular.

Machine learning models, like sequential CNNs and ANNs can also be very helpful in determining the State of Health and Remaining Useful Life (RUL) of a Li-On battery. This data can then be used for analysis and prediction of failure of the battery. The Sequential CNN proposed in paper [16] was able to determine the SoH of the battery with an accuracy of 98.5 %. The loss in prediction in the proposed model was found to be lesser than 0.02 as the number of epochs were increased. Such machine learning algorithms can also be extended to predict and analyse other aspects of the

battery. Machine learning algorithms, in general, tend to make more accurate predictions as they assess a large amount of data and make conclusions from them.

## III. WORKING OF THE PROPOSED SYSTEM

This section briefly explains the workings of the proposed system. The different ways in which battery failure can manifest are as follows:

- Temperature: Extreme temperatures can degrade the battery over time and can impact longevity.
- Overcharging: Overcharging and deep charging can stress the battery and cause premature failure.
- Software issues in the Battery Management System (BMS).
- Age: Li-On Batteries tend to degrade due to internal chemical reactions with time.

The dataset used for this project was taken from Kaggle, which was originally from IEEE data-port [13]. It contains real-time heating and battery data for different driving cycles. The car used for the dataset is a BMW i3 with a 60Ah battery. The dataset is split into different trips, each containing environmental, vehicle, battery and heating circuit data. This data was recorded for 72 driving cycles. The data was also divided into two categories: Category A and Category B. Category A contains vehicle, battery and heating data recorded in summer, and Category B consists of the same data recorded in winter.

The main parameters for predicting battery failure are voltage and current. One prominent way to detect battery failure is to look for any depression in its battery voltage. Temperature also affects the battery hugely. Low temperatures can lead to lesser chemical reactions within the battery, thus reducing its energy and power capacities. Higher ambient temperatures lead to higher internal temperature, which also reduces power capabilities. High temperatures can also lead to thermal runaway, which is one of the reasons for battery destruction and workplace fires. Therefore, temperature must be consistently monitored to detect battery failure in advance.

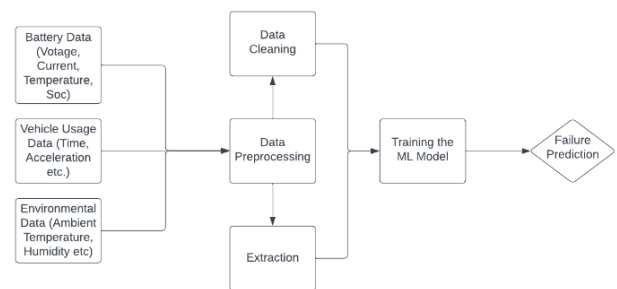


Fig. 1 Block diagram of the proposed model

A schematic block diagram of the developed model and its operation is shown in Fig. 1. The first step is to collect the necessary data. Battery data, vehicle usage data, and environment data are taken with the help of various sensors and then sent for pre-processing. As the dataset may contain data that is not relevant to the model, it requires pre-processing to get rid of such data. Only the required data is extracted and moves on to the next step. Then, the Machine Learning model is trained and deployed to detect any anomalies in the battery's functioning. The Machine learning model accesses the currently measured data and then compares it with the dataset and the ideal characteristics of

the EV. Then, a data-driven conclusion is drawn from the data to predict the chances of battery failure.

#### IV. MACHINE LEARNING MODELS

The dataset consists of 29 columns. The last column, "Failure", shows the chances of failure as high, low or nil.

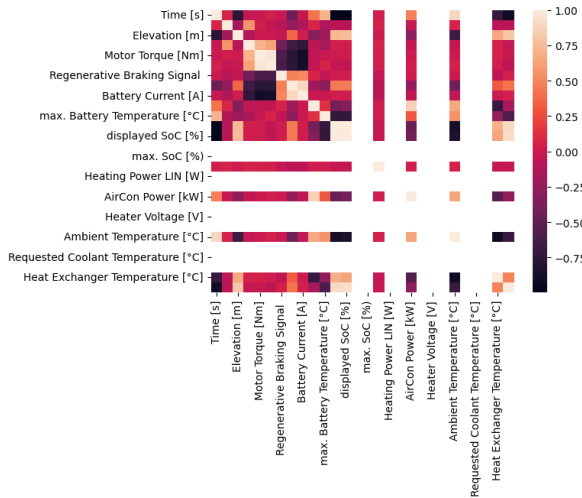


Fig. 2 Correlation table for the dataset

Fig. 2 shows the correlation table for the battery failure prediction dataset. As seen from the figure, most of the columns are highly co-related. For training different machine learning models, the data was split into training and testing data in a ratio of 75:25. The different models used in this work are researched below:

##### A. KNN Regressor

KNN is an algorithm that can act as a classifier and regressor. As this paper focuses on prediction, a KNN regressor has been used.

Mean Squared Error : 0.01801155512758787  
R Squared : 0.8807212397531546  
Mean Absolute Error : 0.04159845931632162

Fig. 3 Error obtained for KNN Regressor

Fig. 3 shows the error values – MSE, MAE and R<sup>2</sup> errors obtained when using the KNN regression algorithm. The R – R-squared value was found to be 0.88, which shows that the model is a good fit for predicting battery failure. It also has very low MSE and RMSE scores, indicating that the deviation of predicted values from actual values is much less.

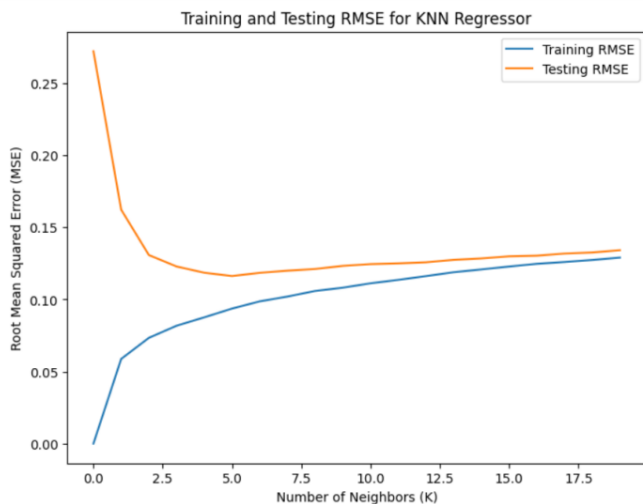


Fig. 4 Training and Testing loss for KNN regressor

The training and testing losses obtained for the KNN regression algorithm are shown in Fig 4. The graph is plotted for K in the range 0 to 20; the error selected for the plot was Root Mean Squared Error (RMSE). It is seen that the testing losses decrease as the numbers of neighbors increase.

##### B. Support Vector Regression (SVR)

Support Vector Regression is an ML algorithm that works on the Support Vector Machine (SVM) principle but is designed for regression/prediction applications instead. SVR was deployed to predict battery failure.

Mean Squared Error : 0.12494293422771166  
R Squared : 0.2113846294496936  
Mean Absolute Error : 0.2814462932577612

Fig. 5 Errors Obtained for SVR

Fig. 5 shows the errors obtained for SVR. The R – R-squared error is 0.28, which is very low and shows low correlation between the data. On the other hand, both MSE and MAE are very low, indicating less deviation between predicted and actual values.

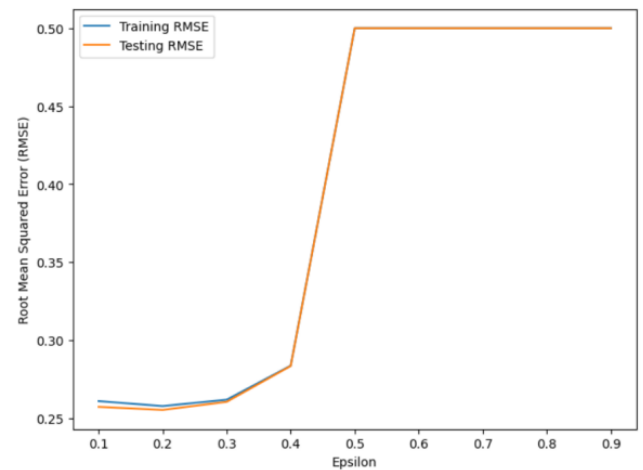


Fig. 6 Training and testing losses for SVR

Fig. 6 shows the training and testing losses obtained for the SVR algorithm. It was plotted for epsilon values ranging from 0 to 0.9, and the selected error was Root Mean Squared Error (RMSE).

##### C. Decision Tree Regressor

Decision Tree (DT) is an algorithm for classification and regression applications. However, the Decision Tree Regressor was deployed as the application involves regression. This algorithm sets decision rules from the data features to predict the target variable.

Mean Squared Error : 0.31150698122291764  
R Squared : -1.0629071873752722  
Mean Absolute Error : 0.31150698122291764

Fig. 7 Errors obtained using Decision Tree regressor

Fig. 7 shows the errors obtained while using the DT algorithm. As shown in the figure, the R – R-squared error is -1.6. A negative R – R-squared error shows that the model underfits or overfits and is unsuitable for the application.

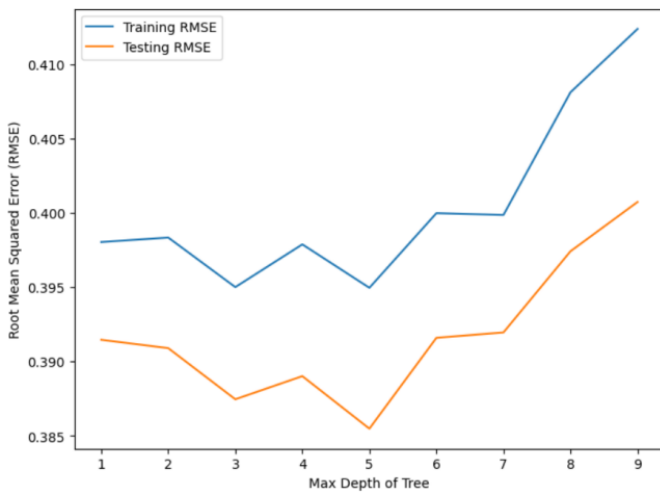


Fig. 8 Training and testing loss of DT algorithm

Fig. 8 shows the plot of training and testing losses for the DT algorithm. The X-axis has the max depth of a tree ranging from 0 to 10, and the Y-axis has the RMSE values corresponding to these depths.

#### D. Random Forest Regressor

Random forest is an algorithm that uses averaging to avoid overfitting and increase prediction accuracy. It uses many decision trees over the dataset's subsamples, thus reducing prediction errors.

Mean Squared Error : 0.0  
R Squared : 1.0  
Mean Absolute Error : 0.0

Fig. 9 Errors obtained for random forest regression  
Training and Testing RMSE for Random Forest Regressor

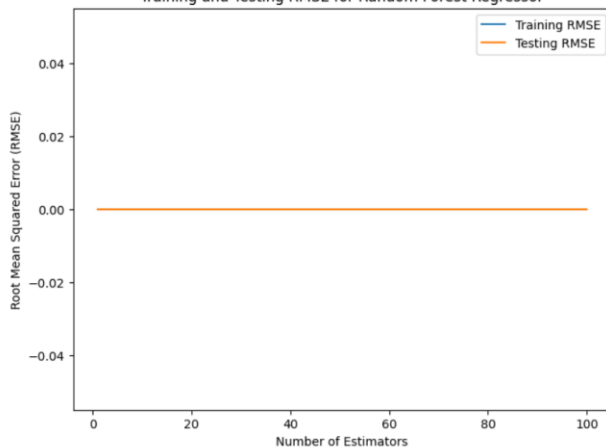


Fig. 10 Training and testing losses for Random Forest Regressor

Fig. 9 shows the errors obtained when the random forest regressor was deployed. The R – R-squared value was 1.00, which shows that the model perfectly fits the application. The MSE and MAE values were also found to be zero, meaning that predicted and actual values are equal.

Fig. 10 shows the training and testing losses obtained for the Random Forest regressor. The x-axis indicates the number of estimators and the y-axis the root mean squared error (RMSE). As can be seen from the figure, both training and testing losses were zero.

#### E. Gradient Boost Regression

It is a machine-learning model that can be used for classification and regression. It uses straightforward decision trees to predict while assuming very little data.

Mean Squared Error : 1.0658547235027836e-10  
R Squared : 0.999999992941541  
Mean Absolute Error : 8.163918432727751e-06

Fig. 11 Errors obtained for Gradient Boost regression

Fig. 11 shows the errors obtained for Gradient Boost algorithms. The MSE and MAE were found to be much less, and the R – R-squared error was almost equal to 1. Hence, the model is almost a perfect fit for predicting battery failure in EVs.

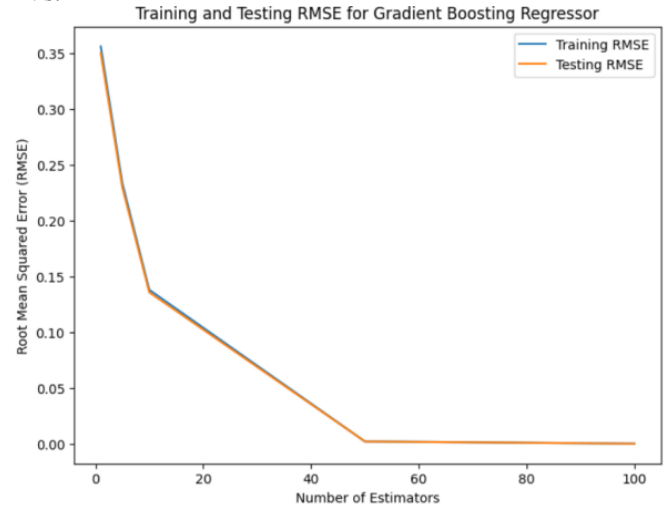


Fig. 12 Training and testing losses for Gradient Boost regressor

Fig. 12 shows the training and testing losses for Gradient boost algorithms. It is a graph plotted between the number of estimators vs Root Mean Squared Error (RMSE). The RMSE values decrease as the number of estimators increase.

#### F. Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is a model based on the neuron networks of the human brain. The layers of ANN are - an input layer, several hidden layers and an output layer. Each layer has nodes and their corresponding weights.

```
Epoch 1/100
195/195 [=====] - 1s 988us/step - loss: 0.5435 - accuracy: 0.7328
Epoch 2/100
195/195 [=====] - 0s 947us/step - loss: 0.3356 - accuracy: 0.8061
Epoch 3/100
195/195 [=====] - 0s 1ms/step - loss: 0.2824 - accuracy: 0.8061
Epoch 4/100
195/195 [=====] - 0s 1ms/step - loss: 0.2469 - accuracy: 0.8185
Epoch 5/100
195/195 [=====] - 0s 1ms/step - loss: 0.2249 - accuracy: 0.9233
Epoch 6/100
195/195 [=====] - 0s 1ms/step - loss: 0.2064 - accuracy: 0.9448
Epoch 7/100
195/195 [=====] - 0s 1ms/step - loss: 0.1868 - accuracy: 0.9559
Epoch 8/100
195/195 [=====] - 0s 1ms/step - loss: 0.1701 - accuracy: 0.9639
Epoch 9/100
195/195 [=====] - 0s 1ms/step - loss: 0.1539 - accuracy: 0.9679
Epoch 10/100
195/195 [=====] - 0s 1ms/step - loss: 0.1275 - accuracy: 0.9711
Epoch 11/100
195/195 [=====] - 0s 1ms/step - loss: 0.0982 - accuracy: 0.9746
Epoch 12/100
195/195 [=====] - 0s 1ms/step - loss: 0.0821 - accuracy: 0.9767
Epoch 13/100
195/195 [=====] - 0s 1ms/step - loss: 0.0702 - accuracy: 0.9788
Epoch 14/100
195/195 [=====] - 0s 1ms/step - loss: 0.0632 - accuracy: 0.9812
Epoch 15/100
195/195 [=====] - 0s 1ms/step - loss: 0.0583 - accuracy: 0.9814
Epoch 16/100
195/195 [=====] - 0s 993us/step - loss: 0.0556 - accuracy: 0.9815
Epoch 17/100
195/195 [=====] - 0s 963us/step - loss: 0.0530 - accuracy: 0.9803
Epoch 18/100
195/195 [=====] - 0s 1ms/step - loss: 0.0504 - accuracy: 0.9817
Epoch 19/100
195/195 [=====] - 0s 1ms/step - loss: 0.0486 - accuracy: 0.9835
Epoch 20/100
195/195 [=====] - 0s 1ms/step - loss: 0.0467 - accuracy: 0.9831
```

Fig. 13 Training of ANN model

Fig. 13 shows the training of the ANN algorithm. The algorithm was trained for 50 epochs. The accuracy for

each epoch is displayed along with the training loss for each epoch. The accuracy was found to be in the range of 95 – 99%.

Mean Squared Error : 0.004619870672426866  
R Squared : 0.9694056153172581  
Mean Absolute Error : 0.011755992072202941

Fig. 14 Errors obtained for the ANN model

Fig. 14 shows the performance parameters of the ANN model. The MSE was calculated to be 0.0046 and the MAE to be 0.011, which are very low. Hence, the predicted values are less deviated from the actual value.

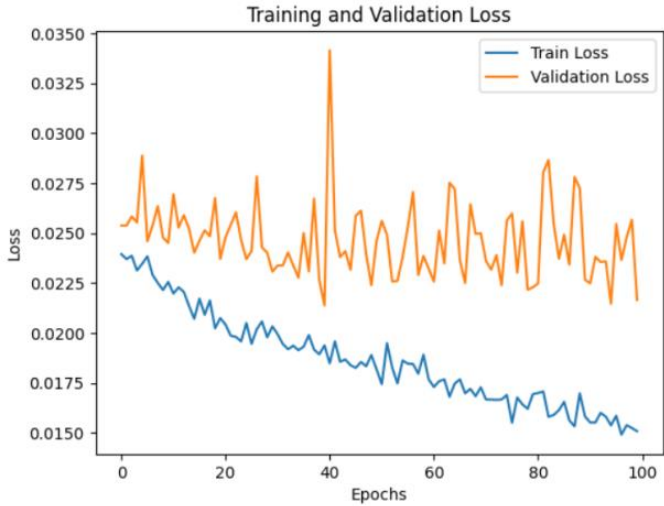


Fig. 15 Training and Validation losses for ANN model

Fig. 15 shows the plot of the training and testing losses obtained for ANN algorithms. The history of the train and test was recorded for 100 epochs. The selected error for this plot was the Mean Absolute Error (MAE), plotted against the number of plots.

G. Long Short Term Memory (LSTM)

LSTM is a Recurrent Neural Network (RNN) type mainly used in sequence prediction applications. It consists of a cell, an input gate, an output gate and a forget gate.

TABLE 1 COMPARISON OF PARAMETERS

Epoch 1/50  
195/195 [=====] - 2s 1ms/step - loss: 0.2930 - accuracy: 0.7203  
Epoch 2/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1483 - accuracy: 0.8878  
Epoch 3/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1254 - accuracy: 0.9116  
Epoch 4/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1145 - accuracy: 0.9218  
Epoch 5/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1090 - accuracy: 0.9259  
Epoch 6/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1059 - accuracy: 0.9275  
Epoch 7/50  
195/195 [=====] - 0s 1ms/step - loss: 0.1016 - accuracy: 0.9334  
Epoch 8/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0982 - accuracy: 0.9364  
Epoch 9/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0959 - accuracy: 0.9334  
Epoch 10/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0934 - accuracy: 0.9392  
Epoch 11/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0907 - accuracy: 0.9372  
Epoch 12/50  
195/195 [=====] - 0s 2ms/step - loss: 0.0883 - accuracy: 0.9421  
Epoch 13/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0867 - accuracy: 0.9416  
Epoch 14/50  
195/195 [=====] - 0s 1ms/step - loss: 0.0854 - accuracy: 0.9433  
Epoch 15/50  
195/195 [=====] - 0s 2ms/step - loss: 0.0834 - accuracy: 0.9458

Fig. 16 Training of LSTM

The LSTM algorithm was trained for 50 epochs, as shown in Fig. 16. The LSTM algorithm's accuracy was between 96 – 97%.

Mean Squared Error : 0.023423547460540817  
R Squared : 0.8448811509120024  
Mean Absolute Error : 0.058661489251088106

Fig. 17 Errors obtained for the LSTM model.

Fig.17 shows the performance parameters of the LSTM model. The MSE was calculated as 0.023, and the MAE as 0.059. The errors are very small in magnitude, and a high R – R-squared value shows that the model is a good fit.



Fig. 18 Training and validation loss of LSTM model.

Fig. 18 shows the training and validation losses obtained for the LSTM model. The history of the train and test was recorded for 100 epochs.

V. RESULT AND ANALYSIS

Various Machine Learning (ML) algorithms were deployed, and their respective performance parameters were recorded using Python code using sci-kit-learn and TensorFlow libraries in Python. The data was scaled to a range (0,1) using the MinMax scaler from scikit-learn. The errors and the plots against selected parameters were drawn and analysed.

Table 1 compares the errors obtained for all Machine Learning (ML) algorithms used in the proposed model. From

| Model          | MSE       | R <sup>2</sup> | MAE       | Accuracy |
|----------------|-----------|----------------|-----------|----------|
| KNN            | 0.018     | 0.88           | 0.416     | ---      |
| SVR            | 0.125     | 0.211          | 0.281     | ---      |
| Decision Tree  | 0.311     | -1.063         | 0.312     | ---      |
| Random Forest  | 0.00      | 1.00           | 0.00      | ---      |
| Gradient Boost | 1.066e-10 | 0.999          | 8.163e-06 | ---      |
| ANN            | 0.0046    | 0.969          | 0.011     | 95 – 99% |
| LSTM           | 0.023     | 0.844          | 0.0587    | 96 - 97% |

the table, it is clear that random forest and gradient boost are the two best-fit algorithms for predicting battery failure in EVs. They both have the least errors, which are very close to zero. Their R - R-squared scores are also almost equal to one, indicating they are a perfect fit for the application. Artificial Neural Networks (ANNs) also provide a high R-squared score and fewer values of MAE and MSE. Thus, ANN is also a good fit but is not preferred over random forest or gradient boost because of its longer computation time and higher

complexity. The computation time of ANN is about 1 minute for 100 epochs, but random forest and gradient boosts usually take less than 10 seconds.

Due to their low MSE and MAE scores, both KNN and Long Short-Term Memory (LSTM) algorithms suit this application. Their R – R-squared values are also around 0.8, showing that they provide a high correlation among data. LSTM has a slightly higher computation time than KNN, around 10 seconds. However, LSTM is a better choice here due to its lower MAE and MSE value. KNN is not preferred because its MAE is too high, indicating heavy deviance from actual values in prediction.

SVR is not a good fit and is not preferred for the application as it has a fairly high MSE and MAE while having a very low R – R-squared value of 0.211. The least preferred ML model is the Decision tree, as it has negative R – R-squared values, showing that the model overfits/underfits. It also has the highest MSE and MAE values, making it the worst choice for predicting battery failure in EVs.

## VI. CONCLUSION

This work deployed different machine-learning models to a dataset with vehicle, environmental, battery and heating circuit data to predict the battery failure in EVs. All the performance parameters of deployed models were analysed, and it can be concluded that random forest regressor and gradient boost regressor are the two best models for this application. They give the least error (MSE and MAE) values and have the highest R-squared values. On the other hand, SVR and decision trees are the least preferred models for this purpose. They have very high error values and very low R-squared values.

ANN and LTSM are also good fits for the application but have higher computation times than the remaining ML models. These models also provide a high accuracy score and can be used for regression and classification. KNN is a decent fit but is still less preferred as it has a very high MAE value, which shows that the predicted and actual values have high deviations.

Machine learning can be used to predict the battery failure of EVs as it provides a prediction of higher accuracy and efficiency. With the increasing importance of electric vehicles, the need for such systems and advanced algorithms has also increased. In the future, advanced algorithms like deep learning models can be integrated to increase accuracy further. IoT and GSM systems can also be used to develop a notification system that can alert the user and schedule maintenance sessions in advance. These systems can be integrated into the battery's BMS. The scope of such systems is very large and will play a major role in the EV industry in the coming years.

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